

7.3 Relationships between variables ↕

□ Watch: Exploring data visually (part 2) [19.42 mins]

Watch the following video, which explains how to construct cross tabulations and their related charts to explore patterns in data. How to create and compare visualisations that display association between quantitative variables is described. Further, correlation is defined and how to estimate its strength is explained.

This is useful as it enables understanding of how to explore association between multiple variables.



Transcript

<https://learn.usc.edu.au/courses/35242/files/3138649?wrap=1>

Source: © University of the Sunshine Coast (2023).

Download the following slides for note taking: [ICT702 Exploring data visually \(part 2\) \(https://learn.usc.edu.au/courses/35242/files/3168023?wrap=1\)](https://learn.usc.edu.au/courses/35242/files/3168023?wrap=1)

After the univariate analysis is complete, the next step is to examine the relationships between two or more variables. This is called *multivariate analysis*. It generally involves pairs of variables, making it bivariate; however, it can involve a greater number of variables. The variables can be categorical or quantitative, with the variable type dictating the statistical summary and visualisation techniques that are used.

□ Watch: Introduction to multivariate analysis [8.22 mins]

This video provides an overview of multivariate analysis and the objectives of examining two or more variables at a time.

As you watch, consider:

- How can multivariate analysis be used to explore and reveal data insights?
- At what stage of data analysis multivariate analysis is important?



Source: © Sneha Jadhav (2020).

Cross tabulation – relationships between categorical variables

Cross tabulations are tables that provide the analytical results of the relationship between two or more (typically categorical) variables. They are simply data tables that show the statistical measures for the variables of interest. They enable the observation of patterns and trends in the data. Although they are most frequently used with categorical variables, they can also be used with quantitative variables by using bins of consecutive values.

□ Watch: What is cross tabulation? [1.58 mins]

This video is useful, as it defines cross tabulation and explains how it is used to find patterns and trends in raw data.

As you watch, consider:

- How could cross tabulation be used to improve the accessibility of the data that you deal with?
- Why could this process be beneficial in exploratory analysis?



Source: © Survey Sparrow (2019).

Pivot tables can be used to examine a cross tabulation of more than two variables. MS Excel offers a visualisation tool called a slicer, which can be used to interactively filter the data by variable.

Association between two quantitative variables

A *scatter chart* enables the analysis of the relationship between two quantitative variables. To convey meaningful patterns well, two elements of the chart must be considered:

- The *aspect ratio* refers to the ratio between the height and the width of the chart. It is important to set the height and the width of the chart the same, as this creates a square chart and does not favour either variable.
- *Quantitative scales* refer to the values on the x and y axes. It is best to set these values to:
 - start at a value slightly lower than the smallest value of each corresponding variable
 - finish on a value slightly higher than the greatest value of each corresponding variable.

A *trendline* can then be added to the scatter chart.

- A *positive slope* (gradually increasing upward from left to right) to a linear trendline indicates a positive correlation between the two variables – i.e. an increase in one variable generally corresponds to an increase in the other variable.
- A *negative slope* (gradually decreasing downward from left to right) to a linear trendline indicates a negative correlation between the two variables.

Correlation is a statistical measure that shows the strength of the linear interdependence between two variable quantities. Correlation values range between -1 and +1. The strength of a correlation can be gauged by the proximity of the data points to the trendline. Values close to zero indicate no evidence for a linear relationship.

A visualisation that indicates a correlation between two variables may also be caused by a *spurious relationship*. A spurious relationship can occur when:

- both variables are impacted by a third variable (called a *lurking variable*)
- the data is biased and not a representative sample
- the data is insufficient to distinguish it from random coincidence.

When examining scatter charts showing pairs of variables, it is important to recognise that a linear trendline will not be universally useful. Using a *nonlinear* trendline in a scatter chart may indicate a relationship between two variables that is not evidenced by a linear trendline.

A *scatter chart matrix* is useful when you want to see pairwise relationships between several different pairs of variables. A scatter chart matrix can show every possible pair of quantitative variables in a dataset in one visualisation.

A *table lens* can also be used to visualise relationships between several different pairs of variables. It consists of a table of data with horizontal bars. The lengths of the bars are proportional to the values in each variable's column. A table lens is a useful technique for visualising wide and tall datasets because insight into the relationships remains visible across different zoom levels. To use a table lens to visualise the relationships between the different variables, the data must be sorted by magnitude (largest values/bars to smallest values/bars) according to one of the quantitative variables. Different pairwise relationships can be analysed by sorting the table on the values of the different variables.

Given that the variable of interest has been sorted from largest value to smallest value, interpreting the meaning of the large bars/small bars pattern across the columns requires the following understanding:

- If there is a *large values to small values* pattern displayed down the related column, it can be deduced that there is a positive association between the two columns/variables.
- If there is a *small values to large values* pattern displayed down the related column, it can be deduced that there is a negative association between the two columns/variables.
- If there is no pattern displayed down the related column, it can be deduced that there is no relationship between the two columns/variables.

□ Watch: Introduction to correlation [4.17 mins]

This video is useful as it provides a simple explanation of correlation.

As you watch, consider:

- How could understanding correlation help you in investigating and telling the story held within a dataset?
- Why is correlation not the same as causation?



Source: © Cody Baldwin (2021).

□ Activity: The world's population density by latitude

Reflect on this webpage about the world's population density in 2022. As you look at the visualisations, consider and make notes on:

- how easy it is for you to comprehend the first visualisation
- the cognitive load for an audience
- how you would approach creating visualisations for the data
- what techniques you would use, and why.

Explain your reasoning and bring your notes to the workshop.

Rae, A., & Lusina, A. (2022, May 6). [The world's population density by latitude](https://www.visualcapitalist.com/cp/mapped-the-worlds-population-density-by-latitude/)  (<https://www.visualcapitalist.com/cp/mapped-the-worlds-population-density-by-latitude/>) . *Visual Capitalist*.

